

Fully Functional Source Code — Overview for Judges

Eco-Loop / TwinPilot · HirePro Honeywell · <https://github.com/alphasafal/Twincity-/tree/ecoloop-hackathon-final>

GitHub (full source)

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Deliverable piece	Path
EnergyPlus wrapper / closed loop	services/simulator/twinpilot_simulator/ep_experiment.py
LLM orchestration (Path C)	scripts/hybrid_supervisory_loop.py, services/agent/
MCP communication bus (stdio)	services/mcp-server/
SafetyShield	services/optimizer/twinpilot_optimizer/safety.py
Savings dashboard	apps/web/ · live https://twinpilot.webyaar.in

Code tour

```
# Code tour – Eco-Loop closed loop (for engineers)

Read this if you need to understand the PoC quickly. Core files are documented with
module/function comments.

## One-picture flow

```text
EnergyPlus Runtime API
 → observations
 → (Path C) MCP stdio client/server + Ollama strategy
 → deterministic optimiser / agent proposal
 → SafetyShield validate
 → set_actuator_value(Clg-SetP-Sch)
 → next EnergyPlus state + logs
...

Entry points (start here)

| Role | File | What to read first |
|-----|-----|-----|
| EnergyPlus experiment |
[`services/simulator/twinpilot_simulator/ep_experiment.py`]
(../services/simulator/twinpilot_simulator/ep_experiment.py) | `run_experiment`,
`propose_agent_cooling_setpoint`, `validate_setpoint_action` |
```

```
| API SafetyShield | [`services/optimizer/twinpilot_optimizer/safety.py`]
(../services/optimizer/twinpilot_optimizer/safety.py) | `SafetyShield.validate` |
| MCP server tools | [`services/mcp-
server/twinpilot_mcp/energyplus_experiment_server.py`](../services/mcp-
server/twinpilot_mcp/energyplus_experiment_server.py) |
`build_energyplus_experiment_fastmcp` tool list |
| MCP stdio client | [`services/mcp-server/twinpilot_mcp/stdio_session.py`]
(../services/mcp-server/twinpilot_mcp/stdio_session.py) | `StdioMcpSession`,
`open_energyplus_mcp_session` |
| Path C hybrid | [`scripts/hybrid_supervisory_loop.py`]
(../scripts/hybrid_supervisory_loop.py) | `call_ollama_strategy`,
`setpoint_from_strategy`, `make_hybrid_proposal_fn` |
| Path A scripts | [`scripts/run_baseline.sh`](../scripts/run_baseline.sh),
[`scripts/run_agent.sh`](../scripts/run_agent.sh) | shell → `ep_experiment` |
| Dashboard data | API building status + `DATA_MODE=energyplus` | reads `results/*`
only |
```

## How to reproduce

```
```bash
./scripts/run_baseline.sh && ./scripts/run_agent.sh && ./scripts/compare_results.sh
./scripts/run_hybrid_supervisory_experiment.sh
DATA_MODE=energyplus ./scripts/run_demo.sh
```
```

## Claims mapping

- **\*\*Savings / comfort numbers\*\*** → Path A (`results/comparison/`, `final-release/evidence/independent-metrics.json`)
- **\*\*LLM + MCP agency\*\*** → Path C (`results/hybrid/`, MCP traces)
- **\*\*Safety\*\*** → `safety-rejection.log` (35°C) + `ollama-fallback.log`